MITRE ATLAS | AML.CS0000 - Evasion of Deep Learning Detector for Malware C&C Traffic

Entry Information

ID:
AML.CS0000

Type:
Case-Study

Name:
Evasion of Deep Learning Detector for Malware C&C Traffic

URL:
https://atlas.mitre.org/studies/AML.CS0000/

Incident Date:
2020-01-01

Target:
Palo Alto Networks malware detection system

Actor:
Palo Alto Networks AI Research Team

Description

The Palo Alto Networks Security AI research team tested a deep learning model for malware command and control (C&C) traffic detection in HTTP traffic.
Based on the publicly available [paper by Le et al.](https://arxiv.org/abs/1802.03162), we built a model that was trained on a similar dataset as our production model and had similar performance.
Then we crafted adversarial samples, queried the model, and adjusted the adversarial sample accordingly until the model was evaded.

Attack Procedure:
[AML.T0000.001] Pre-Print Repositories: We identified a machine learning based approach to malicious URL detection as a representative approach and potential target from the paper [URLNet: Learning a URL representation with deep learning for malicious URL detection](https://arxiv.org/abs/1802.03162), which was found on arXiv (a pre-print repository).
[AML.T0002.000] Datasets: We acquired a command and control HTTP traffic dataset consisting of approximately 33 million benign and 27 million malicious HTTP packet headers.
[AML.T0005] Create Proxy AI Model: We trained a model on the HTTP traffic dataset to use as a proxy for the target model.
Evaluation showed a true positive rate of ~ 99% and false positive rate of ~ 0.01%, on average.
Testing the model with a HTTP packet header from known malware command and control traffic samples was detected as malicious with high confidence (> 99%).
[AML.T0043.003] Manual Modification: We crafted evasion samples by removing fields from packet header which are typically not used for C&C communication (e.g. cache-control, connection, etc.).
[AML.T0042] Verify Attack: We queried the model with our adversarial examples and adjusted them until the model was evaded.
[AML.T0015] Evade AI Model: With the crafted samples, we performed online evasion of the ML-based spyware detection model.
The crafted packets were identified as benign with > 80% confidence.
This evaluation demonstrates that adversaries are able to bypass advanced ML detection techniques, by

crafting samples that are misclassified by an ML model.

References

- Le, Hung, et al. "URLNet: Learning a URL representation with deep learning for malicious URL detection." arXiv preprint arXiv:1802.03162 (2018). — https://arxiv.org/abs/1802.03162